TECHNICAL GEOLOGY & RESOURCE REPORT / 02

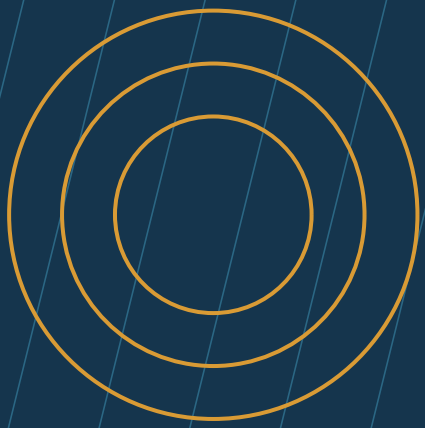
SILICAFLUX

Mineral silica: global occurrence, high-purity quartz and where to find it

MINERAL RESOURCE INTELLIGENCE

FROM NATURAL QUARTZ TO A SEMICONDUCTOR FEEDSTOCK

Geology / Occurrence / Global localities / Exploration priorities



Purpose

Identify the principal geological forms of natural silica, distinguish broad industrial resources from high-purity quartz (HPQ), and prioritise global districts for SilicaFlux exploration and supply-chain development.

Prepared for	Technical focus	Report date
SilicaFlux development programme	Natural quartz and silica resources for silicon and semiconductor supply chains	2 November 2026

Use limitation

This is a desktop geological screening report. Named districts are exploration leads or known industrial localities, not automatically economic deposits or semiconductor-qualified feedstock. No mineral resource or reserve is declared. Ownership, title, permitting, access, environmental constraints and current operating status require separate verification.

Executive summary

Silica is abundant; semiconductor-suitable quartz is not. The mineral quartz (SiO_2) occurs in igneous, metamorphic and sedimentary rocks, in hydrothermal veins, pegmatites, quartzites, cherts, sandstones and unconsolidated sands. Most natural silica is unsuitable for the highest-value applications without upgrading. The commercial question is not simply “where is quartz?” but “where is quartz with a favourable impurity inventory, texture, tonnage, continuity and route to market?” [1–4]

For SilicaFlux, the most relevant targets are coarse-grained pegmatite and hydrothermal-vein quartz, followed by selected sedimentary/chert and quartzite deposits where impurities can be liberated economically. The best documented benchmark remains the Spruce Pine district in North Carolina. Norway has a developed quartz industry and several recognised deposit types. Australia has one operating high-purity silica-to-silicon chain at Moora, Western Australia, plus newer state and national prospectivity work. Canadian and Chinese districts offer additional targets, but their products and project claims must be evaluated deposit by deposit. [3–10]

The distinction between purity categories is essential. USGS describes HPQ as quartz with less than 100 ppm total impurities (at least 99.99% SiO_2); ultra-high-purity materials may contain 10 ppm or less. These are mineral/product descriptors, not a finished semiconductor silicon specification. Silicon intended for semiconductor devices is chemically purified after reduction, and crystal and wafer qualification impose a separate set of controls. [1,3]

Strategic finding	Implication for SilicaFlux
Quartz is widespread; high-value HPQ deposits are rare.	Do not screen by SiO_2 assay alone. Geology, trace elements and impurity occurrence determine processability.
Pegmatite and hydrothermal-vein settings dominate the leading HPQ examples.	Prioritise simple, coarse quartz bodies with low Al, Ti, B, P and Fe; test fluid and mineral inclusions.
Australia and Norway provide strong strategic jurisdictions and public geoscience data.	Build a first-pass portfolio around Moora / NSW and Norwegian districts; use Spruce Pine as a benchmark.
Industrial silica sand is not automatically semiconductor-grade quartz.	Treat UK silica sand as a local industrial feedstock opportunity unless advanced assays prove otherwise.

Recommendation: assemble a tiered exploration portfolio. First, benchmark against commercial HPQ; second, assay known high-potential quartz districts; third, only fund bulk samples and beneficiation pilots after trace-element and petrographic gates are passed.

Contents and scope

01 What silica is
02 Geological settings and deposit types
03 Global locations and prospectivity
04 United Kingdom resource picture
05 How to identify a worthwhile deposit
06 SilicaFlux exploration priorities
07 Risks, economics and development pathway
08 Sources and technical notes

Scope note

This report focuses on natural silica minerals and rocks that might supply quartz concentrate or silica feedstock for silicon metal and downstream semiconductor materials. It does not attempt to catalogue every silica occurrence, every sand producer or every industrial silica product. Localities are grouped by evidence strength: established commercial supply, documented deposit or exploration project, and regional reconnaissance potential. A “high-purity” label in a company or government report is not equivalent to a customer-qualified HPQ product.

01 / WHAT THE MINERAL IS

Silica is silicon dioxide, SiO_2. In nature it occurs as the mineral quartz, other crystalline polymorphs, microcrystalline forms and amorphous silica. Silicon itself is not normally found as native metal in the Earth's crust; it is chemically combined with oxygen in silica and silicate minerals. Quartz is the principal natural mineral source used in the silicon production chain. [1,11]

The main natural forms

Form	Description	Relevance to SilicaFlux
Quartz	Crystalline SiO_2; common in veins, pegmatites, granites, quartzites and sediments.	**Primary target.** The preferred natural mineral feedstock for silica and HPQ evaluation.
Chert / flint / chalcedony	Microcrystalline or cryptocrystalline silica, commonly sedimentary or replacement origin.	Potential feedstock where bulk impurity, texture and beneficiation behaviour are favourable; Noondine Chert in WA is a notable silicon-chain example.
Cristobalite / tridymite	Less common natural SiO_2 polymorphs; also form in high-temperature settings and industrial products.	Not the normal exploration target for silicon metal; quartz is the dominant natural route.
Opal / amorphous silica	Non-crystalline hydrated or poorly ordered silica, including biogenic and weathering-derived forms.	Important in other markets, but not a first-choice feedstock for crystalline silicon reduction.
Silicate minerals	Silica combined with other elements in feldspar, mica, pyroxene and many other minerals.	Not free quartz; high-grade silica recovery from complex silicates is usually a different and more difficult processing proposition.

Useful physical properties of quartz

Quartz has Mohs hardness 7 and specific gravity around 2.65. It is chemically robust and occurs in many colours and textures. A white, glassy or massive quartz outcrop is only a visual clue; colour and appearance cannot establish low trace-element content. Geological Survey of Québec describes quartz as the most abundant natural crystalline form of silica and lists quartz, cristobalite and tridymite as polymorphs. [11]

Purity terminology

General industrial silica — Usually application-specific; industrial sands often >95% SiO2.

Silicon-metal furnace feed — Commonly around 98-99% SiO2 (deposit and furnace dependent).

High-purity quartz (USGS definition) — At least 99.99% SiO2 / below 100 ppm total impurities.

Ultra-high-purity quartz — 10 ppm or less total impurities in demanding applications.

Purity must be interpreted against the intended product. SiO_2 weight percentage is only one metric; semiconductor and fused-quartz customers also specify individual trace elements, inclusions, thermal behaviour, grain size, and batch consistency. The assay of a raw ore, a beneficiated concentrate, a fused-quartz product and electronic-grade elemental silicon are four different things. [1,3,12]

02 / GEOLOGICAL SETTINGS

HPQ deposits form where quartz was concentrated and impurity introduction or retention was limited. The same broad geological type can host both excellent and poor material. A favourable district is therefore a search model, not a grade guarantee. Recent work links deposit genesis, impurity occurrence and the best purification pathway; coarse liberatable gangue is generally easier to upgrade than lattice-bound contamination or fine intergrowths. [12,13]

01

PEGMATITE

Coarse quartz + feldspar / mica; crystal liberation can be favourable.

02

HYDROTHERMAL VEIN

Quartz precipitated from fluids; inspect fluid inclusions and trace elements.

03

QUARTZITE

Large-tonnage recrystallised sandstone; grain boundaries may lock impurities.

04

SEDIMENTARY

Quartz sand, chert, pebbles; coatings, clays and heavy minerals matter.

Pegmatites

Granitic pegmatites are exceptionally coarse-grained igneous bodies, often composed mainly of quartz and feldspar with mica and accessory minerals. Large quartz crystals may be liberated by selective mining and crushing. The best targets are relatively simple, rare-metal-poor pegmatites with low concentrations of tourmaline, beryl, apatite, rutile, garnet and other impurity-bearing phases. Complex Li-Cs-Ta pegmatites can be mineralogically zoned: some quartz may be attractive, but accessory minerals and fluid evolution can create variable impurities. Spruce Pine, Drag and East Qinling are relevant examples. [5,6,9,13]

Hydrothermal quartz veins

Quartz precipitates from silica-bearing fluids moving through fractures and shear zones. Very large, massive veins can contain high silica and low visible gangue. The key risks are fluid inclusions, microfractures, wall-rock contamination and trace elements incorporated during crystal growth. Vein material may need controlled thermal pretreatment to open or decrepitate inclusions before chemical cleaning. Norway’s Nasafjell and Nesodden are named exploration examples. [5,12,13]

Quartzite and quartzitic sandstone

Quartz-rich sandstone recrystallised by metamorphism can produce large, laterally extensive quartzite bodies. These may offer scale and simple mining geometry, but interlocking grain boundaries, mica-rich seams, feldspar, iron oxides and fine inclusions can make liberation difficult. Tana in Norway and the Mount Wilson Formation in British Columbia are important industrial examples. Grade suitable for silicon-metal production does not by itself demonstrate HPQ suitability. [5,8,10]

Sedimentary quartz, chert and alluvial accumulations

Weathering, transport and sorting can remove some less-resistant minerals and concentrate quartz grains or pebbles. Silica-rich sands, chert and quartz pebble horizons may be attractive because large-volume sedimentary processes can create extensive deposits. Risks include clay coatings, iron staining, heavy-mineral contamination and variable grain size. Australia’s Noondine Chert and Cowra/Glenella quartz pebbles are important examples. [8,4]

Impurity occurrence is the deciding variable

Occurrence	Typical examples	Processing implication
Surface / grain coatings	Iron oxides, clay, alkali salts, weathering products	Often accessible by washing, scrubbing, attrition, magnetic separation or leaching.
Discrete mineral inclusions	Feldspar, mica, rutile, tourmaline, hematite, zircon	Liberation, sizing, flotation and high-gradient magnetic separation; success depends on grain size.
Fluid inclusions / microfractures	Water, CO_2 and dissolved metal salts in healed cracks	May require thermal decrepitation or controlled cracking followed by cleaning.
Lattice-bound impurities	Al, Ti and other elements within quartz crystal defects	Most difficult to remove; may disqualify a deposit from high-end use.

03 / GLOBAL LOCATIONS & PROSPECTIVITY

The following shortlist separates known commercial districts from emerging prospects. Locations are not ranked by tonnage alone. Evidence status, impurity data, beneficiation history, infrastructure and jurisdiction all matter.

North America

Spruce Pine district, North Carolina, United States — commercial benchmark. The district lies in the Blue Ridge Province and is known for pegmatite/alaskite bodies. Sibelco states that its IOTA high-purity quartz products are mined from two ore bodies at Spruce Pine and are used in fused-quartz components for semiconductor and photovoltaic applications. USGS records two US HPQ producers around Spruce Pine in its 2025 quartz summary. This is the strongest benchmark for natural HPQ quality and process performance, but access to new resource or partnership opportunities may be constrained by incumbent operators and established supply chains. [6,14,15]

British Columbia, Canada — Mount Wilson Formation / Moberly and Nicholson. The Ordovician Mount Wilson quartzite belt near Golden, including Moberly and Nicholson, is a large industrial silica source. British Columbia's 2024 mining overview reports Moberly silica at about 99% SiO_2 and historic stockpiles / contract sales; that is useful industrial-grade material, but it is not proof of semiconductor HPQ. The Mount Wilson belt remains relevant for large-volume silica supply and further trace-element screening. [16]

Québec, Canada — Baie-Johan-Beetz / Crête White. A project promoter describes a quartz/silica pegmatite property about 13 km east of Baie-Johan-Beetz and reports approximately 5.8 Mt exposed at surface across two principal veins. The project page gives coordinates of 50°17′ N, 62°38′ W and describes quartz pegmatite bodies formed during metamorphism and recrystallisation. Treat the tonnage and quality as company-reported project information pending independent technical due diligence and current resource classification. [17]

Europe

Drag, Nordland, Norway — commercial HPQ processing district. Norway's Geological Survey states that The Quartz Corp extracts and processes local and imported HPQ at Drag, near Narvik. Pegmatites in the Drag region have been periodically produced alongside other southern Norwegian pegmatite districts. This is a meaningful European benchmark with an established processing ecosystem. [5]

Setesdal and southwest Norway — pegmatite prospectivity. Setesdal and surrounding areas have a history of intermittent quartz production; NGU reports additional southwest-coast pegmatites with interesting HPQ qualities. These are geological leads requiring deposit-specific assay and process testing. [5]

Nasafjell and Nesodden — hydrothermal vein exploration. NGU identifies the Nasafjell veins near the Arctic Circle and the Nesodden deposit by the Hardangerfjord as large high-grade veins being explored for possible mining. They should be tracked as exploration projects, not treated as producing HPQ mines. [5]

Tana, Mårnes and Kragerø — quartzite / quartzitic sandstone. These are established Norwegian quartz quarry areas, particularly important for ferrosilicon and other silicon-material supply. NGU states that Tana material is used in an advanced refinement route to produce high-grade silicon metal; this is distinct from proving that all of the run-of-mine quartzite is HPQ. [5]

Sweden and France. European technical literature identifies hydrothermal-vein and aggregate sources in these countries used in the wider high-purity quartz supply chain. Individual orebody and product specifications should be confirmed with producers before using these as exploration targets. [18]

03 / GLOBAL LOCATIONS & PROSPECTIVITY (CONTINUED)

Australia and Oceania

Moora, Western Australia — Noondine Chert / operating silicon chain. Geoscience Australia's 2024 review identifies Simcoa in Western Australia as Australia's only HPQ mining and silicon-manufacturing operation at that time. Its feedstock is the Noondine Chert near Moora, a Neoproterozoic siliceous unit. This is the most commercially relevant Australian silica-to-silicon reference point and a priority location for benchmarking ore characteristics, beneficiation requirements and downstream integration. [8,19]

New England Orogen, New South Wales — Kingsgate, Deepwater and Tenterfield. The Geological Survey of NSW evaluated more than 105 silica occurrences and completed 404 geochemical analyses. Fifty samples exceeded 99.8% SiO_2, the highest reported assay was 99.96%, and ten samples also had low measured impurity levels below 500 ppm. The Kingsgate, Deepwater and Tenterfield districts returned the state's top-tier purity results. These are promising exploration areas, not yet an established HPQ mine or semiconductor-qualified feedstock. [4]

Other NSW targets. The Curnamona Province near Broken Hill (Thackaringa and Euriowie pegmatites), Lachlan Orogen hydrothermal veins (Caloola and Rockley), and Cowra/Glenella sedimentary quartz pebble deposits are also identified by the NSW study. Their deposit styles differ, so SilicaFlux should not use one universal beneficiation flowsheet. [4]

Queensland and national Australian prospectivity. Geoscience Australia's 2026 High Purity Silica Mineral Systems Study provides a continent-scale mineral-potential assessment and an economic-fairways approach. Mount Isa–Cloncurry is one region under investigation. These are modelled prospective areas and exploration frameworks, not measured HPQ resources. [3,20]

Asia

East Qinling, Henan, China — emerging pegmatite research district. Recent 2026 research on a granitic pegmatite quartz sample reports that beneficiation followed by chlorination roasting reduced total impurity concentration to 17.26 micrograms/g and produced a nominal SiO_2 purity of 99.9982%. The authors describe this as potential photovoltaic-grade HPQ sand. The result demonstrates processing potential for the tested material; it is not a mine-wide grade, reserve or commercial production specification. [9]

Other Chinese districts. A 2026 review identifies the Eastern Qinling Mountains (Henan), Altay (Xinjiang), Dabie Mountains (Hubei), and selected parts of Jiangxi and Sichuan as important quartz-resource regions. It also emphasises strong internal variability and uneven ore quality. [10]

India and Sri Lanka. Published reviews identify both countries as participants in the wider HPQ supply base. At project level, grade, trace elements, mineral texture, and downstream product qualification must be checked independently. [8,14]

South America, Africa and the Middle East

Brazil. Quartz resources and mines are widely reported in Pará, Minas Gerais, Santa Catarina and Bahia. These are significant quartz-bearing provinces and worthwhile regional exploration areas, but country-level quartz abundance should not be confused with a defined semiconductor-grade HPQ resource. Historical literature notes Brazil's former role in lump-quartz exports. [8,14]

Angola, Mauritania, Namibia, Cameroon and Madagascar. Regional reviews identify these countries as having geological potential for HPQ-style pegmatites, hydrothermal veins or quartzites, but also describe limited production, processing infrastructure or public project definition. Treat them as frontier reconnaissance jurisdictions rather than supply-ready sources. [8,10]

Saudi Arabian Shield. Hydrothermal quartz veins have been identified as a prospective silica style in the Arabian Shield. The proposition requires detailed field sampling, trace-element work and infrastructure assessment before any ranking as a development project. [8]

04 / UNITED KINGDOM RESOURCE PICTURE

The UK has a well-established silica-sand industry, but most documented deposits are developed for glass, foundry, filtration, sports and other industrial applications. They are useful for local silica supply and could provide low-cost screening samples; the public data reviewed here do not establish a UK source of semiconductor-grade HPQ. [21]

Important British silica-sand districts

Region	Known resource / context	SilicaFlux interpretation
Lochaline, Morvern, Scotland	Lochaline White Sandstone Formation; BGS factsheet describes it as the highest-purity silica sand in Britain and reports Fe_2O_3 typically below 0.0085%. Underground operation has a long history.	High-quality industrial silica benchmark. Test full trace-element suite before assuming HPQ suitability.
Cheshire	Chelford and Congleton sands; Pleistocene glaciofluvial deposits. BGS describes them as important UK production sources.	Large industrial-sand resource; geological variability and historic depletion of some units must be checked.
Norfolk	Leziate Member, Lower Cretaceous quartz sands near King's Lynn; BGS documents uses in colourless glass, flat glass and foundry sand.	Good source of industrial silica; not automatically HPQ.
Surrey, Kent, Bedfordshire, Essex, North Lincolnshire	Lower Cretaceous sands and other silica-sand deposits; BGS lists multiple established production areas.	Potential industrial feedstocks and process trials; grade must be established deposit by deposit.
Nottinghamshire, North Yorkshire, Worcestershire, Hampshire	Additional regional producers / deposits in the BGS Mineral Planning Factsheet.	Consider where logistics or local industrial uses favour supply; no semiconductor HPQ assumption.
Fife, West Lothian, North Ayrshire, Highland	Scotland has silica-sand operations including Devilla Forest, Fife, and Lochaline, Morvern.	Local bulk and specialist silica opportunities; check present operator and planning status.

The British Geological Survey / Mineral Products Association factsheet documents silica sand producers in Cheshire, Surrey, Norfolk, North Lincolnshire, Bedfordshire, Nottinghamshire, North Yorkshire, Kent, Essex and other areas, as well as sites in Scotland. It is a 2020 factsheet drawing on earlier production surveys, so the named operator list and site status must be refreshed before investment or offtake discussions. [21]

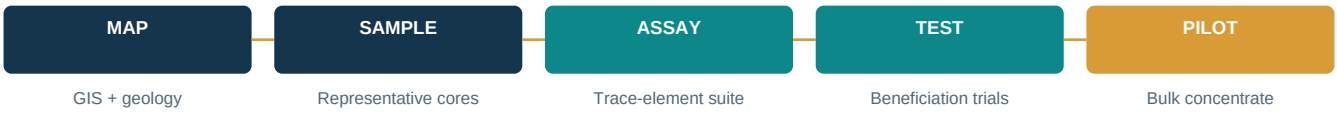
UK recommendation

For a UK-based SilicaFlux programme, Lochaline is the strongest first domestic case study for premium industrial silica, while Cheshire and Norfolk provide broader silica-sand comparators. None should be labelled semiconductor HPQ solely on the basis of BGS descriptions or Fe_2O_3 content. The correct first step is to obtain representative bulk samples and commission full ICP-MS / LA-ICP-MS impurity analysis plus beneficiation trials.

05 / HOW TO IDENTIFY A WORTHWHILE DEPOSIT

The deposit must be assessed as an integrated mineral-processing system. A high SiO_2 assay can conceal high aluminium, titanium, boron or phosphorus, and a deposit with clean chemistry can still be uneconomic if quartz is intergrown with gangue or locked in fine textures. The 2026 geological-genesis / impurity-occurrence / purification framework recommends screening lattice potential, inclusion accessibility and liberation constraints in sequence. [12]

A staged exploration and test-work programme



Workstream	Recommended methods	Decision question
Geological mapping	Outcrop mapping, structural mapping, intrusive / vein relations, unit continuity, regolith and overburden mapping.	Is there a coherent body with mineable geometry and consistent quartz mineralisation?
Petrography / mineralogy	Thin sections, reflected / transmitted light, XRD, SEM-EDS, cathodoluminescence where useful.	What minerals and inclusion types are present, and how are they distributed?
Trace-element chemistry	LA-ICP-MS spot analyses; whole-rock ICP-MS after validated preparation; GDMS or equivalent for very low concentrations.	What are the levels and variability of Al, Ti, Fe, B, P, Li, Na, K, Ca and other buyer-critical elements?
Texture and liberation	Size-by-size mineralogy, fracture mapping, inclusion mapping, liberation analysis.	Can impurities be physically separated without excessive fine grinding or yield loss?
Beneficiation tests	Scrubbing, classification, magnetic separation, flotation, thermal treatment and carefully controlled leaching.	Which flowsheet removes which impurity, at what mass yield and cost?
Product qualification	Bulk concentrate trials, customer samples, crucible / fused-silica or silicon-smelting tests as relevant.	Does the material pass a written customer specification consistently?

Critical analytical cautions

Use representative channel or drill-core samples, not hand-picked white quartz. Separate sampling should cover the full body, alteration margins, weathered zones, internal pegmatite zones and visible accessory-mineral concentrations. Analyse both head grade and the beneficiated concentrate. Trace-element techniques must be selected for the required detection limits; bulk ICP-MS alone can dilute grain-scale hotspots, while LA-ICP-MS reveals within-crystal variation. Include blanks, certified reference materials, duplicates and umpire laboratory checks.

Priority impurity suite

Priority	Why it matters	Likely mineral / source relationship
Al, Ti	Can be difficult to reduce to very low levels; lattice and inclusion forms matter.	Feldspar, mica, rutile, titanite, lattice substitution or micro-inclusions.
Fe	Impacts colour and process quality; often removable if surface-bound or in liberated oxides.	Hematite, goethite, pyrite, iron-bearing silicates and coatings.
B, P	Important electrically active contaminants in silicon routes; must be assessed at very low levels.	Tourmaline / apatite and related accessory minerals; trace incorporation.
Li, Na, K, Ca, Mg	Application-specific alkali and alkaline-earth impurity limits.	Mica, feldspar, fluid inclusions, carbonate or other gangue.

06 / SILICAFLUX EXPLORATION PRIORITIES

A practical portfolio should mix one global benchmark, a small number of well-documented development opportunities, and low-cost regional screening. Ranking below is an internal desk-study judgement based on published evidence, not a formal prospectivity model.

Priority	Locality	Evidence class	Why it matters	Key diligence gap
A	Spruce Pine, North Carolina, USA	Commercial HPQ benchmark	Established HPQ feedstock / high-spec fused-quartz supply chain.	Access, incumbent concentration, offtake and orebody-level trace chemistry.
A	Moora / Noondine Chert, Western Australia	Operating silica-to-silicon chain (reported 2024)	Relevant Australian mine-to-silicon model and local industrial capability.	Current operations, grade variability, cost and product specification.
A-	Drag, Norway	Commercial HPQ processing district	European benchmark and established processing know-how.	Local vs imported feed mix, current capacity and resource access.
B+	Kingsgate–Deepwater–Tenterfield, NSW	Government geochemical prospectivity	Top-tier NSW assay results from broad survey.	Drill-defined continuity, full trace-element suite, bulk yield.
B	East Qinling, Henan, China	Recent laboratory beneficiation evidence	Recent study achieved 99.9982% nominal SiO_2 product from a sample.	Representative deposit scale, consistency, commercial process performance.
B	Baie-Johan-Beetz, Québec	Promoter-defined project	Reported exposed quartz veins and accessible coastal location.	Independent technical report, validated resource, product test work.
B-	Glenella / Cowra and other NSW targets	Government-identified sedimentary target	Pebble quartz source and potential tonnage.	Full chemistry and mineral liberation.
C+	Lochaline, Scotland	Established industrial silica sand	High-grade British silica sand comparator; local supply chain.	Not demonstrated as semiconductor HPQ; complete trace suite required.
C	Brazil, Angola, Mauritania, Namibia and other frontier belts	Regional geological potential	Quartz-rich districts and potential pegmatite / vein systems.	Country-level data are insufficient; require named deposits and local sampling.

Portfolio logic

The first investment should not be a new mine. It should be a feedstock qualification programme: acquire representative samples from several candidate districts, run the same analytical suite, and compare purification response, mass yield, impurity rejection and cost. This produces comparable evidence and prevents a single high headline assay from dominating the decision.

Recommended SilicaFlux target specification development

SilicaFlux should not publish a single universal “HPQ specification” before talking to intended customers. Instead, create separate feedstock envelopes for (1) metallurgical silicon furnace quartz, (2) fused-quartz / crucible-grade quartz, and (3) high-end optical / semiconductor quartzware. Each envelope should identify maximum individual impurities, particle size, inclusion limits, thermal stability and permitted variability. Final elemental silicon and wafer specs are downstream products and require separate qualification. [1,3,12,18]

07 / FIELD PROSPECTING PLAYBOOK

Where to look first

At regional scale, compile geological maps, mineral occurrence databases, geochemical surveys, geophysical interpretations, historic mine data, satellite imagery and land-tenure constraints. Target the contact zones of evolved granites and pegmatite fields, major brittle structures with quartz-vein systems, mature metamorphic belts containing quartzite, and selected sedimentary units with extensive quartz pebble or chert horizons.

Field indicators worth following

Positive indicator	Caution / false positive
Large, coarse, pale quartz bodies with relatively simple mineral assemblages.	White colour and transparency do not prove low trace elements.
Pegmatite with abundant liberated quartz and limited complex rare-metal accessories.	Some highly fractionated pegmatites can concentrate Li, Cs, Ta, B and phosphate minerals that degrade quartz purity.
Massive hydrothermal vein with low visible wall-rock breccia and low inclusion density.	Fluid inclusions and lattice elements may persist even when the vein looks clean.
Quartzite or chert with uniform chemistry and low non-quartz mineral content.	High SiO_2 can coexist with difficult intergranular locking, very fine grain size or variable beds.
Consistent assays across multiple representative samples and depth intervals.	A single exceptional grab sample can be misleading and cannot support a resource estimate.

Field sample protocol

For each target, take spatially distributed channel or core samples across the body, log structures and alteration, record GPS and sample interval, photograph the in-situ exposure, and preserve duplicates. Avoid sampling only the cleanest-looking quartz. Record host rock, accessory minerals, fractures, weathering, grain size, outcrop continuity and overburden. A preliminary programme can use a limited number of representative samples per geological domain; the exact number should be set by deposit variability and the decision being made.

Bulk test-work ladder

Gate	Indicative sample scale	Pass condition
Reconnaissance	Hand specimen to tens of kg	No obvious disqualifying mineralogy; initial chemistry justifies further work.
Bench beneficiation	Tens to hundreds of kg	Repeatable impurity reduction and acceptable concentrate yield.
Pilot	Hundreds of kg to tonnes	Stable flowsheet, water and reagent balance, product consistency.
Pre-feasibility	Representative bulk sample and drilling	Mineable geometry, resource confidence, infrastructure, permitting and offtake case.

Environmental and health considerations

Mining and crushing quartz can generate respirable crystalline silica dust. Dry drilling, crushing, screening and transfer points require engineered dust suppression or extraction, enclosure, monitoring and appropriate respiratory protection. Chemical purification may use corrosive acids, oxidants or chlorination chemistry and needs closed handling, scrubbers, effluent treatment, secondary containment and formal process-safety review. Water demand, wastewater salts/metals, tailings, biodiversity, land rights and community consent must be assessed early. A clean mineral does not automatically create a clean or low-cost process.

08 / CONCLUSIONS & ACTION PLAN

SilicaFlux should approach silica as a mineral systems and processability problem, not a commodity tonnage problem. Quartz is widespread, but HPQ-grade ore is restricted to unusual geological histories and favourable impurity distributions. The most compelling exploration targets are pegmatites and hydrothermal veins with low lattice impurities and accessible inclusions; selected chert, sedimentary quartz and quartzites can provide alternatives where their texture and chemistry allow economical upgrading.

Recommended next 90 days

Work package	Action	Output
1. Build the data room	Compile public geological maps, occurrence data, historical assays, title and land access for benchmark and prospect districts.	GIS prospect database and initial target map.
2. Define SilicaFlux assay suite	Set detection limits and QA/QC for SiO_2, Al, Ti, Fe, B, P, Li, Na, K, Ca, Mg, plus mineralogical and inclusion work.	Sampling and laboratory protocol.
3. Acquire representative samples	Seek legally obtained samples from Spruce Pine benchmark products, Moora/WA, Norway, NSW and Canadian project sources where available.	Comparative sample set; chain-of-custody records.
4. Run comparative bench tests	Test physical liberation, magnetic separation, flotation, thermal pretreatment and controlled leaching on selected samples.	Mass-balance and impurity-rejection curves.
5. Select one pilot target	Rank grade variability, yield, unit costs, access, energy, water, permitting and offtake.	Go / no-go decision for bulk sampling and pilot plant.

Bottom line

Best immediate benchmark: Spruce Pine, North Carolina. **Best integrated Australian reference:** Moora / Noondine Chert. **Best near-term exploration shortlist:** Kingsgate–Deepwater–Tenterfield in NSW and Norwegian pegmatite / vein districts. **Most interesting emerging processability case:** East Qinling, China, where 2026 laboratory work reported a high-purity concentrate from pegmatite quartz. **Best UK comparator:** Lochaline, Scotland, with the caveat that published evidence supports premium industrial silica sand, not proven semiconductor HPQ.

SilicaFlux should only designate a deposit “semiconductor-relevant HPQ” after representative drilling, trace-element assays, mineralogical characterisation, beneficiation mass balance and customer qualification.

SOURCES & TECHNICAL REFERENCES

Sources were consulted on 2 November 2026. The report prioritises geological surveys, government technical publications, peer-reviewed studies and clearly attributed company project descriptions. Where a company-reported resource or product claim is used, it is labelled as such.

[1] U.S. Geological Survey. “Silica Statistics and Information.” Commodity scope and HPQ definition (<100 ppm total impurities; ultra-high-purity may be at most 10 ppm).
https://www.usgs.gov/centers/national-minerals-information-center/silica-statistics-and-information

[2] U.S. Geological Survey. Mineral Commodity Summaries 2026. Industrial sand and gravel chapter; 2025 estimated mine production and global market context.
https://pubs.usgs.gov/publication/mcs2026

[3] Geoscience Australia. “High Purity Silica,” Australian Critical Minerals R&D; Hub; page updated 4 September 2026.
https://www.ga.gov.au/scientific-topics/minerals/australian-critical-minerals-research-and-development-hub/high-purity-silica-mineral-systems-study

[4] Geological Survey of New South Wales. “High-purity silica project.” Study of >105 occurrences and 404 geochemical analyses; results and target districts.
https://www.resources.nsw.gov.au/geological-survey/projects/high-purity-silica-project

[5] Geological Survey of Norway (NGU). “Quartz resources in Norway.” Deposit types, production districts and exploration targets.
https://www.ngu.no/en/geological-resources/quartz-resources-norway

[6] Sibelco. “High Purity Quartz.” IOTA HPQ source ore bodies at Spruce Pine and semiconductor / crucible applications.
https://www.sibelco.com/en/materials/high-purity-quartz

[7] U.S. Geological Survey. “Geology of the Spruce Pine district, Avery, Mitchell, and Yancey Counties, North Carolina.” Bulletin 1122-A (1962).
https://www.usgs.gov/publications/geology-spruce-pine-district-avery-mitchell-and-yancey-counties-north-carolina

[8] Jennings, A. et al. “A review of high-purity quartz for silicon production in Australia.” Australian Journal of Earth Sciences (2024).
https://doi.org/10.1080/08120099.2024.2362296

[9] Yi, Y. et al. “Purification of Quartz from East Qinling Granitic Pegmatite for the Preparation of High-Purity Quartz Sand.” Minerals 16(9), 938 (13 September 2026).
https://doi.org/10.3390/min16090938

[10] Zhang, H. et al. “Global and Chinese distribution of quartz resources and the strategic value of China’s quartz endowment.” Chinese Journal of Engineering, 48(5), 1010–1021 (2026).
https://doi.org/10.13374/j.issn2095-9389.2025.09.09.006

[11] Gouvernement du Québec, Géologie Québec. “Quartz (Silice).” Mineral properties, geology, exploration and extraction in Québec.
https://gq.mines.gouv.qc.ca/portail-substances-minerales/la-silice/

[12] Guo, F. et al. “The interplay among geological genesis, impurity occurrence, and purification of high-purity quartz ores.” Minerals Engineering 241 (1 July 2026), 110181.
https://doi.org/10.1016/j.mineng.2026.110181

[13] Shao, H. et al. “Review of advanced purification technologies for high-purity quartz materials.” Materials Research Bulletin 196 (March 2026), 113904.
https://doi.org/10.1016/j.materresbull.2025.113904

[14] U.S. Geological Survey. Mineral Commodity Summaries 2025, “Quartz (High-Purity and Industrial Cultured Crystal).” US HPQ producers around Spruce Pine.
https://pubs.usgs.gov/periodicals/mcs2025/mcs2025-quartz.pdf

[15] Geoscience Australia. “New insights into Australia’s unknown potential for high-purity quartz resources.” 26 August 2024.
https://www.ga.gov.au/news/new-insights-into-australias-unknown-potential-for-high-purity-quartz-resources

[16] British Columbia Geological Survey. Provincial Overview of Exploration and Mining in British Columbia, 2024; Moberly silica mine / Mount Wilson Formation.
https://minedocs.com/28/British-Columbia-Geological-Survey-2024-Information-Circular-012025.pdf

[17] PAL Quartz. “Discover Baie-Johan-Beetz Deposit / Crête White Property.” Company project description and promoter-reported tonnage.
https://www.palquartz.com/project

[18] “Perspectives for High-Purity Quartz from European Resources.” Minerals 15(10), 1080 (2025). Europe HPQ producers and resource context.
https://www.mdpi.com/2075-163X/15/10/1080

[19] Geoscience Australia. “A review of high-purity quartz for silicon production in Australia” summary; Moora / Simcoa / Noondine Chert.
https://www.ga.gov.au/news/new-insights-into-australias-unknown-potential-for-high-purity-quartz-resources

[20] Geoscience Australia. “National-scale mineral potential assessment of high purity silica mineral systems in Australia” (2026) and related map/data links.
https://www.ga.gov.au/scientific-topics/minerals/australian-critical-minerals-research-and-development-hub/high-purity-silica-mineral-systems-study

[21] British Geological Survey / Mineral Products Association. “Mineral Planning Factsheet: Silica sand” (January 2020). UK resource areas, Lochaline and industry context.
https://www.mineralproducts.org/MPA/media/root/Publications/2019/mpf_silica_sand.pdf

Interpretation and data-quality notes

“High-purity quartz,” “high-purity silica,” “silica sand,” “quartzite” and “silicon feedstock” are not interchangeable terms. This report uses HPQ for high-grade natural quartz products, with the USGS threshold as a broad reference only. Individual customer specifications may be tighter or differ by application. Exploration target rankings are qualitative. Company-reported figures are not presented as independently verified resources. Historical production lists and operating status should be checked with regulators and operators before commercial reliance.

Prepared as a SilicaFlux desk-study report. It is not a competent-person resource statement, feasibility study, environmental impact assessment or mining / process design package.